Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1 manual

Course Number: 30206111

"قـــرار رقـم ٩-٢٠١٩/٢٠٢٠"

التنسيب باعتماد أدلة التجارب للمواد العملية في قسم الفيزياء والعلوم الاساسية

١-	الفيزياء العامة العملي ١
٢-	الفيزياء العامة العملي ٢
٣-	الكيمياء العامة العملية ١
٤-	مختبر كيمياء عضويه
٥-	مختبر تحليل الي
٦-	مختبر كيمياء تحليلية

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جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية

تعليمات السلامة العامة في مختبرات الكيمياء والهندسة الكيميائية

1	منع دخول من ليس له عمل رسمي إلى المختبر.
2	يجب الإلتزام والتقيد بمواعيد بداية وانتهاء محاضرات المختبر.
3	يُمنع نهائيا تناول الطعام والشراب والتدخين داخل المختبر.
4	يلتزم كل طالب بطاولة العمل التي يعينها له مشرف المختبر.
5	يجب إرتداء مريول العمل ومعدات السلامة العامة داخل المختبر.
6	يجب وضع الملابس والحقائب في المكان المخصص داخل المختبر.
7	يمنع نهائيا تشغيل الأجهزة إلا بعلم مشرف المختبر.
8	يُمنع نهائيا ترك التجربة خلال تنفيذها بدون مراقبة.
9	يجب التعرف إلى أماكن وجود معدات السلامة العامة داخل المختبر، والإستخدام الصحيح لها.
10	إتباع تعليمات مشرف المختبر وقراءة التعليمات قبل بدء التجربة.
11	يجب الالتزام باستخدام شفاط الأبخرة أثناء العمل بأي تجربة تتضمن أبخرة سوائل متطايرة، وحسب التعليمات.
12	التعامل الحذر مع معدات وأجهزة المختبر على اختلاف أنواعها، والتعامل بجدية تامة داخل المختبر.
13	التعامل بحذر مع المواد الكيميائية المتطايرة والسائلة و الصلبة منها على حد سواء.
14	الحذر عند استخدام اللهب والمعدات المرافقة له، والتأكد من إغلاقه محكما عند الإنتهاء من التجربة.
15	الحذر عند إستخدام الأجهزة الكهربائية والتأكد من فصلها عن التيار الكهربائي عند الإنتهاء منها.
16	يُمنع سكب المواد الكيميائية السائلة في شبكة الصرف الصحي، وإنما يتم تفريغها في الإناء المخصص .
17	يجب مراعاة شروط السلامة العامة في تخزين المواد الكيميائية وحفظها في الأماكن المخصصة لها.
18	الإلتزام التام بطرق حمل الأدوات والعبوات الزجاجية، والحذر عند العمل بها.
19	على الطلبة اللذين يعانون من ظرف صحي معين (حساسية، حمل،...)، إبلاغ مشرف المختبر بذلك.
20	إبلاغ مسؤولي المختبر مباشرة عن: تحطم الأدوات، انسكاب المواد، والحالات الطارئة.
21	التزام الهدوء والاتزان والنظام والإنضباط والنظافة العامة داخل المختبر.
22	التجاوب الفوري مع التعليمات السريعة والاستثنائية التى يعطيها مشرف المختبر واستعمال سرعة البديهة والعقلانية فى تقدير الموقف للمغادرة أو البقاء عند حدوث الخطر -لا قدر الله - وينبغى أن يتم ذلك بدون أي إرباك أو فوضى.

Experiment No.	Experiment Name	Pages
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Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 1

Safety Rules

Student Name:

Student Number:

Submission Date:

Experiment No. 1

Safety Rules



The following safety rules must be followed at all times in the laboratory.

Be sure that you understand every instruction before proceeding. Never hesitate to ask questions especially if there is any question concerning proper operating procedure.

General Rules:

- First of all, you should know the location and the proper use of all safety equipment in your laboratory including:

Fire extinguishers, **fire blankets**, **safety showers**, **eye wash devices**, and **first aid kits.**

Safety Equipment	Name	Use
	**Fire extinguishers**	An active fire protection device used to extinguish or control small fires
	**Fire blankets**	A safety device designed to extinguish incipient (starting) fires.

	**Safety showers**	A unit designed to wash an individual's head and body which has come into contact with hazardous chemicals.
	**Eye wash devices**	An eyewash station is a unit for washing chemicals or substances that might splash into an individual's eyes before he or she can seek further medical attention
	**First aid kits**	A collection of **supplies** and equipment that is used to give medical treatment.

- Read the entire experiment before coming to laboratory.
- Lab coats and eye googles must be worn in the laboratory all the time.
- Eating, smoking, and drinking are not allowed in a chemistry laboratory.
- Do not put any personal belongings on the benchtop.
- Using cell phone or ear phone is not allowed during experimental work.
- Inform the instructor about any accident happening during your work immediately.

Chemicals:

- Every chemical should be treated as though it were dangerous.
- When handling any toxic or hazardous agent, always wear the appropriate gloves.
- Always keep chemicals in their proper place. Do not carry original containers of chemicals (stock bottles) to your benchtop.

- Before using any chemical, carefully read the label on the reagent bottles.

- Never taste or smell any chemical. Some chemicals are very corrosive and poisonous even in very small quantities.

- Avoid getting chemical into contact with your eyes or your skin. Some reagents are irritating and corrosive, if you do, rinse immediately with large quantities of water and inform your instructor immediately.

- Learn the pictograms that appear on the labelling of chemical bottles, indicating their hazardous effect. **Figure (1)**. Pictograms are symbols used worldwide to alert users to chemical hazards to which they may be exposed. Some chemicals are very toxic like methanol, some are flammable (for example, ethanol, methanol, benzene) and others are corrosive like concentrated acids and bases (ex, NaOH ,HCl). In general, chemical materials should be used with great caution.

 **Flammable**	 **Explosive**	 **Toxic**	 **Corrosive**
 **Oxidizing**	 **Dangerous to the environment**	 **Harmful/Irritant**	 **Radioactive**

Figure (1): Pictograms showing hazardous effect of chemical compounds.

- Do not return unused chemicals back into their original stock container. Therefore, one should take very small amounts of the chemicals especially in the beginning.

- Never pour water into concentrated acid but pour acid slowly into water.

- Use the **fume hood** when toxic or irritating vapors are involved.

- Concentrated acids and bases are stored in the fume hood. Do not carry them to your bench.

- Mouth suction is never used to fill a pipette. Use a pipette filler.

COMMON LABORATORY APPARATUS

1) Balances

A balance is an instrument that measures the mass of body. Different types of balances are used in chemical laboratories.

A. **Top loading balance** (with accuracy ± 0.01) shown in **Figure (2-a).**

B. **Analytical balance** (with accuracy ± 0.0001) shown in **Figure (2-b).**



Figure (2-a) Top loading balance



Figure (2-b). Analytical balance

The following rules summarize the operation of weighing objects:

- Select a suitable container for weighing (paper, beaker)
- Place the container on the balance pan, zero the balance by pressing **tare** button.
- **Carefully** transfer the desired chemical to the container. Use a clean dry spatula.
- Clean any chemical spilled on the balance pan, if spill occurs clean it up immediately.
- If you added more than the desired amount, do not return the excess amount back to the bottle, just put it in another container for further use.
- Record the balance reading in your notebook.

Note: For accurate mass measurements, all objects must be weighed at room temperature

2) Glassware:

Laboratory glassware is equipment made of glass used for scientific experiments in chemistry labs and have a wide variety of functions including; storing materials, measuring volumes, mixing and preparation of solutions, heating, cooling, or as containers for chemical reactions. Table 1 shows the most common glassware used in chemistry labs.

Table (1): Laboratory glassware

Glassware	Name	Use
	**Beaker**	Beakers are useful as a reaction container, and for holding liquid or solid samples, mixing, and heating chemicals.
	**Erlenmeyer flask** **(Conical flask)**	Conical flasks are useful as reaction containers or to hold liquid samples, they are easier for mixing than beakers. Flasks are also useful to perform titration experiments.
	**Volumetric flask**	Used for preparation of a solution with a certain concentration.
	**Graduated cylinder**	Used for measurement of an amount of liquid. The volume of liquid can be estimated to the nearest 0.1 mL with practice.
	**Pipette**	Pipettes are used to dispense small quantities of liquids. (Reads to nearest 0.01)

	**Burette**	Used are for addition of a precise volume of liquid. Usually used for titrations. The volume of liquid can be estimated to the nearest 0.01 mL with practice.
	**Round bottom flask**	It is a flask with a round body and flat bottom. Used to hold and heat liquids.
	**Desiccator**	Common use for desiccators is to protect chemicals which are hygroscopic or which react with water from humidity.
	**Test tube**	Used for holding small samples or to mix a small amount of liquid or aqueous chemical(s).
	**Funnel**	Used for funneling liquids from one container to another (ex. filling a burette with a solution) or for filtration.
	**Glass rod**	Used to mix or stir liquids and chemicals

3) Bunsen Burner:

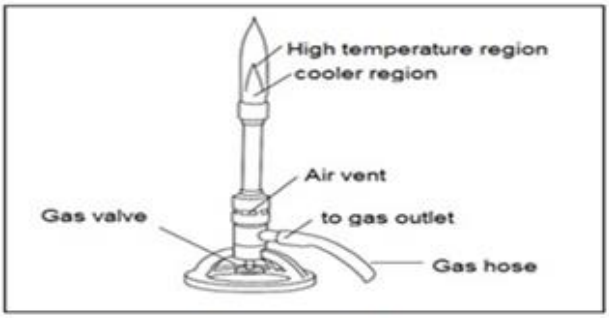


Figure (3) Bunsen Burner

Figure (3) shows the Bunsen burner (named for Robert Bunsen), a common laboratory equipment used for heating, and combustion. The gas used can be natural gas (which is methane) or a liquefied petroleum gas such as a mixture of propane and butane). Always make sure that the rubber tubing is connected to the Bunsen burner before using.

Adjust the height of the flame using the gas valve at the bottom of the burner. For a clean burning flame (complete combustion of the gas), adjust the air vent, to produce a blue flame with no yellow tip. Yellow flames are considered dirty due to incomplete combustion of the gas which leaves a layer of carbon on heating objects.

Using Fire:

- Heating flammable chemicals should be carried out in water baths. Never heat any chemical over a direct flame.

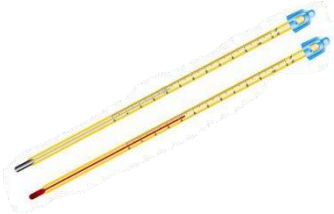
- When using a Bunsen Burner be careful to place it away from any overhead shelving, or other equipment. Remove all papers, notebooks, combustible materials and excess chemicals from the area.

- Do not leave open flames unattended (i.e. one should keep watching the flame) and never leave laboratory while burner is on. Shut off gas valve when burner use is completed.

- Thoroughly wash your hands after leaving the laboratory, and make sure your work area is clean and dry. Ensure that all gas, water, vacuum, and air valves are completely turned off.

4) Other common laboratory apparatus

Table (2) shows other common equipment used in your experiments.

 **Crucible** A small cup equipment made of porcelain or nonreactive material used to heat chemical compounds to very high temperatures.	 **Wash bottle** A squeeze bottle with nozzle (filled with distilled water) usually used for rinsing glassware.	 **Fire gauze** Helps to protect the glassware heated directly on Bunsen burner**.**
 **Thermometer** Used to measure the temperature of objects.	 **Stand with rings**	 **Clamps** Used to hold beaker or flasks during heating process.
 **Dropper** Used to transfer small quantities (few drops) of liquids	 **Tongs** Used to pick up hot objects.	 **Clay triangle** Used in process of heating substances by Bunsen burner
 Filter paper Used to separate solid precipitates from liquids.	 **Spatula** Stainless steel equipment used to collect small quantities of a chemical material	 **Test tube holder**

5) Reagent Bottles

Reagent bottles are containers made of **glass** (for liquid materials) **or plastic** (for solid or powder materials) **Figure (4)**.

To avoid any contamination of chemical material, never put anything back into the reagent bottle.

A certain amount of liquid can be obtained in several ways. For **liquid materials**, pour from the reagent bottle into a beaker and draw up the liquid into a pipet from the beaker.

For solids, pour the solid into a beaker, or onto a weighing paper. Never pour excess back into the bottle. Reagents are to remain on the plastic on the center bench.



Figure (4)

Reading volumetric measurements

In all volumetric glassware (pipet, burette, volumetric flasks, graduated cylinder, etc.), it is necessary to read the level of a liquid. A liquid in a small-diameter container will form a curve at the surface of the liquid. curved surface downward to a minimum at the center. To read the level of the liquid properly, the eye should be at the same level as the bottom of the curve.

Figure (5) shows the correct volume reading for both a cylinder and burette equipment.

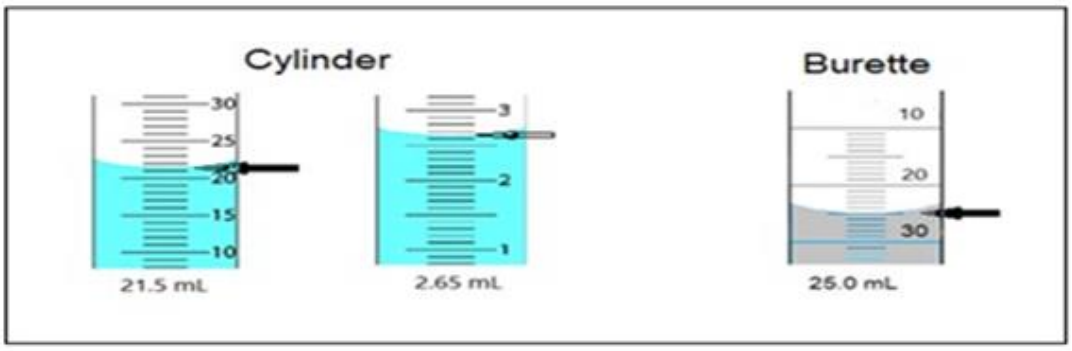


Figure (5)

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Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 2

Physical Properties: part 1 (Density and Solubility)

Student Name:

Student Number:

Submission Date:

Experiment No. 2: Physical properties (I)

Solubility and Density

Objectives

1) To investigate the solubility of various compounds in set of solvents.
2) To determine the density of unknown solid & liquid.

Physical property is the one in which there is no change in the chemical composition but, it represents a transformation from one state to another. For example, the formula of water in solid, liquid or gas state is H_2O. This structure is remained as it is, even if we melt and evaporate water. The formula of water remains constant during physical changes.

The physical properties include;

Solubility

The solubility is defined as the amount of material that dissolves in a given amount of solvent. The solubility of any compound depends on many factors which can be summarized as follows:

- The type of solute and solvent including polarity the solvent and solute. The general accepted rule is ***(likes dissolve likes)***. For example, the length of carbon chains in organic compound especially containing and the presence of high electronegative atom affects the type of bonding between the molecules (intermolecular attractive forces).

- The temperature. Generally, the solubility increases by increasing the temperature.

Notice that the pressure does not affect the solubility of solids or liquids, but it increases the solubility of gases in liquids. This means a positive relationship between them.

- Solubility in water:

Soluble Compounds	**Insoluble Exceptions**
Group 1 metals and NH_4^+	
Nitrates (NO_3^-) and acetates (CH_3COO^-)	
Chlorides (Cl), bromides (Br^-) and iodides (I^-)	Ag^+, Pb^{2+} and Hg^+
Sulfates (SO_4^{2-})	Ag^+, Pb^{2+}, Hg^+, Ba^{2+}, Sr^{2+} and Ca^{2+}
Insoluble Compounds	**Soluble Exceptions**
Carbonates (CO_3^{2-}), sulfites (SO_2^{2-}) and phosphates (PO_4^{3-})	Group 1 metals and NH_4^+
Hydroxides (OH^-)	Group 1 metals, NH_4^+, Ca^{2+} and Ba^{2+}
Sulfides (S^{2-})	Group 1 and 2 metals and NH_4^+
Oxides (O^{2-})	Group 1 metals, NH_4^+, Ca^{2+} and Ba^{2+}

Chemicals & Equipment:

-Various inorganic and organic compounds as NaCl, Na_2CO_3, benzoic acid, ethanol,…..etc.
- Various solvents as water, hexane, acetone …etc.
-Test tube & test tube holder.
-Spatula.

Procedure

1) Introduce few crystals of the solid material in a clean and dry test tube, and then add about 2.0 ml of the solvent. Shake well and record your observation.

2) For liquid materials, place about 1.0 ml of the liquid in a clean dry test tube. Then add about 3.0 ml of solvent. Shake a well and then record your observation.

3) Record your results in the solubility table by using the terms; soluble, insoluble, slightly soluble.

Density:

It is defined as the mass of a substance per unit volume.

Since pure substances have unique density values, measuring the density of a substance can help identify that substance. Density is determined by dividing the mass of a substance by its volume:

$$Density = \frac{Mass}{Volume}$$

densities are reported (g/ml) or (g/cm^3) (*which are equivalent because 1ml ≡ 1cm^3*).
Experimentally, mass and volume measurements are required to calculate density.
Masses are measured on balances. Volume is an amount of space, and it will be measured in many ways, but the units are usually milliliters (mL) or cubic centimeters (cm^3). Methods for determining or delivering precise volumes include volumetric flask and pycnometers; less precise methods include graduated cylinders.
In this experiment, you will measure masses and volumes to determine density.

Part A: the density of an irregular shaped solid.

Materials & equipment:

-Graduated cylinder to measure approximate volume, pycnometers, volumetric flask.
-Various irregular metals such as copper, zinc, aluminum ,…etc.
-Various liquids. -Digital balance. – Water. – Dropper.

Procedure

- Weighing an unknown solid by using balance.
- Put certain amount of water inside the cylinder and record the initial volume.
- Hold the cylinder at 45° angle and slide the metal into the water gently to avoid breaking the bottom of the cylinder.
- Record the final volume of water.

The difference in volume will be the volume of the metal.as in Figure (6)

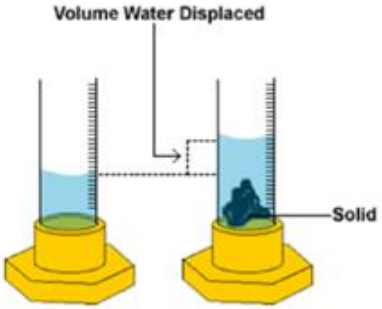


Figure (6)

Part B: density of unknown liquid.

Procedure:

1- Weigh an empty volumetric flask (50 ml) or a pycnometer (as Figure 7) with known volume.
2- Fill the volumetric flask/ pycnometer with the unknown liquid exactly to the mark, (use dropper to reach).
3- Weigh the volumetric flask /pycnometer with content.



Figure (7)

Data & Calculation:

Solubility:

*The table below for testing solute with solvents:

Solute / **Solvent**				

Part A: the density of an irregular shaped solid:

Unknown NO.:_____

Mass of the solid	
Initial volume of water	
Final volume of water	
Volume of solid	
Density of solid	

Show Calculation:

Part B: density of unknown liquid.

Unknown NO.:_____

Mass of empty volumetric flask / Pycnometer	
Mass of volumetric flask / Pycnometer + liquid	
Mass the unknown liquid	
Volume of unknown liquid	
Density of liquid	

Show Calculation:

Questions

1) Which of the following compounds is water soluble? Why?
   A) K_2SO_4
   B) C_2H_5OH
   C) $CH_3(CH_2)_{10}CH_3$
   D) $CaCO_3$

2) In which of the following solvents KCL is dissolve? Explain your answer.
   A) H_2SO_4
   B) HCl
   C) C_6H_6 Benzene

Explanation:

3) What is the affect if you introduced a large quantity of compound into test tube during the experiment of solubility?

4) Will the measured density of a solid material be affected if a part of the solid dissolved in the liquid used and the volume increased?

Explanation:

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 3 Physical Properties: part 2 (Melting and Boiling Points)

Student Name:

Student Number:

Submission Date:

Experiment No. 3: Physical Properties (Part II)

Melting and Boiling Point

Objectives

A) To identify an unknown solid by measuring its melting point. (M.P)

B) To identify an unknown liquid by measuring its boiling point. (B.P)

A) Melting point:

It is defined as the temperature at which crystalline solid change to liquid or melts. At melting point both solid and liquid phases of the substance are in equilibrium.

Factors affecting the melting point:

a) Substance purity affects the range of melting point. This means; if the substance is melted at a sharp point (or has a narrow melting range), then the substance has high purity. However, if it has a wide range (more than 2 ˚C) then the substance is not pure.
b) Nature of the impurities. For example, ionic compound existed in sample the range of melting points increase.
c) The nature of attractive forces between the substance molecules.

Material & Equipment:

- Melting point apparatus. Figures (8-a)
- Thermometer.
- Capillary tubes & unknown sample. Figure (8-b)



Figure (8-a)

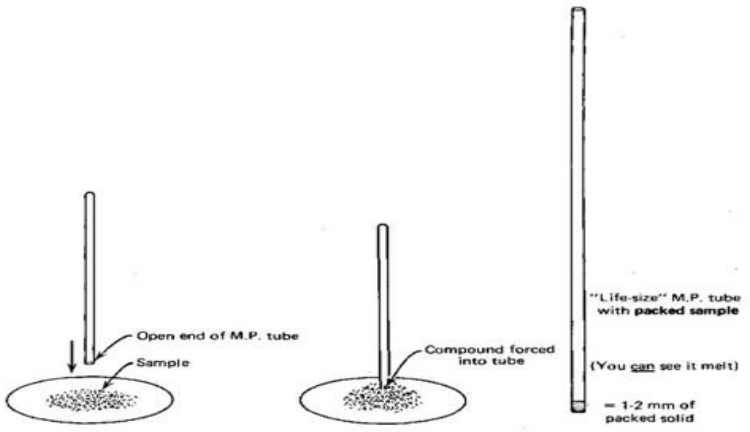


Figure (8-b)

Procedure

a) Introduce about (0.3 – 0.5 mm) dry solid of the unknown sample in a capillary tube with one closed end and pack them well.
b) Place the capillary tube in the melting point apparatus.
c) Adjust the rate of heating temperature rises only one to two degrees per minute.
d) Record the melting point ranges observed.

Identify your unknown by a comparison between your recorded result and a list of melting points of known substances.

B) Boiling point:

It is defined as the temperature at which vapor pressure of liquid is equal to the external pressure. Also, it is the temperature at which both liquid and vapor phases are in equilibrium.

Factor affecting the boiling point:

a- External atmospheric pressure. The relationship is positive with boiling point.

b- The nature of attractive forces between the substance molecules. For example, substances with H-bonding forces have higher boiling point than those having London forces.

c- The molar mass of the substance (directly proportional).
d- Presence

Material & Equipment

- Thermometer - Small test tube - Capillary tube - Rubber band
- Water path (Beaker filled with water) - Tripod stand - Wire gauze
- Unknown liquid - clamps

Procedure

a- Fix the test tube with thermometer by using rubber band to be confirmed that test tube and thermometer bulb at the same level.
b- Add about 1.0 ml from the unknown liquid in test tube.
c- Dipped the capillary tube as reverse position.
d- Set the boiling point apparatus as shown in figure (9).
e- Heat apparatus over water bath gently until you notice the evolution acceleration of air bubbles from below the capillary tube.

f- Turn the burner of & record the thermometer reading.

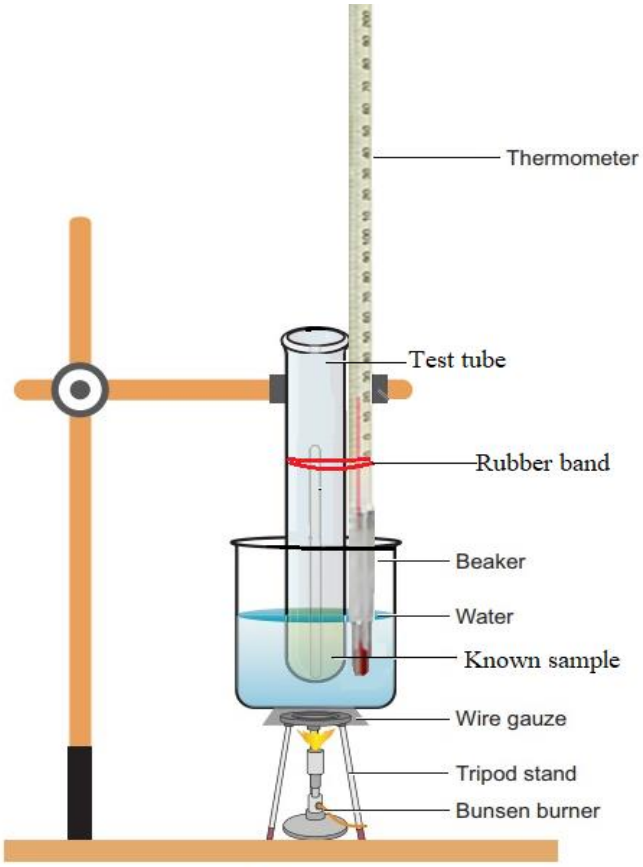


Figure (9)

Data & Results:

The Melting point of unknown: ----------

The Boiling point of unknown: ----------

Questions

1) What is the effect of each following factors on the measured melting point of a solid (increase, decrease, has no affect) & give good explanation for each answer?

a- Rapid heating:

b- A large quantity of compound was introduced into the capillary tube:

c- Impurities:

2) What is the affect of each following factors on the measured boiling point of a liquid (increase, decrease, has no affect)?

a- If the test tube base and the thermometer bulb are not at the same level:

b- If there's some glass shards with liquid sample:

c- If the atmospheric pressure increase:

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General chemistry lab 1

Course Number: 30206111

Experiment No. 4

Formula of Hydrated Salts

Student Name:

Student Number:

Submission Date:

Experimental No. 4:

Water in a Hydrated Salt

Objectives

1. To determine the percent (by mass) of water in an unknown hydrated salt.

2. To determine the number of water molecules per formula unit of the unknown hydrated salt.

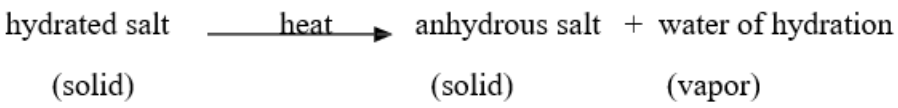
Introduction

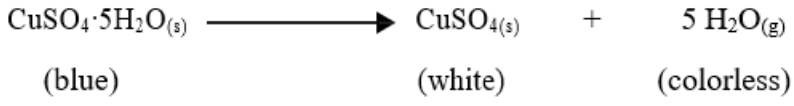
Many salts occurring in nature or purchased from chemical suppliers are hydrated; that is a number of water molecules are bound to the ions in the crystalline structure of the salt. The number of moles of water per mole of a particular hydrated salt is usually a constant. For example, ferric chloride is purchased as $FeCl_3.6H_2O$, not as $FeCl_3$; cupric sulphate as $CuSO_4.5H_2O$, not as $CuSO_4$.

For example, $CuSO_4 \cdot 5H_2O$ means that 5 water molecules are in chemical combination with each formula unit of $CuSO_4$. Formulas and names of a few common hydrates follow:

- $CuSO_4 \cdot 5H_2O$: copper (II) sulfate pentahydrate.
- $BaCl_2 \cdot 2H_2O$: barium chloride dihydrate.
- $Na_2CO_3 \cdot 10H_2O$: sodium carbonate decahydrate.

The stability of hydrates varies. Some spontaneously lose their water molecules and others must be heated to high temperature to drive off the water of hydration. Substances which absorb water from the atmosphere are called hygroscopic (for example $CuSO_4$). Some hygroscopic substances (for example $CaCl_2$) absorb so much water that they form solutions. The hydrates used in this experiment require some heat to drive off the water of hydration, as following:





Some indication help use to know that water molecule is drive off such as that the color may change or salt may have expanded and frothed.

Salts that contain water as part of their crystal structure are called hydrates (or hydrated salts) and the water in the crystal structure is called the water of hydration (or water of crystallization). When the water of hydration is removed from the hydrate, the salt that remains is anhydrous.

The percent of water in a hydrate can be determined experimentally, a hydrate is heated to a constant mass. The decrease in mass of the sample is due to the amount of water lost, and the remaining material is the anhydrous salt.

- mass of water lost = mass of hydrated salt - mass of anhydrous salt
- $\%\ \text{water} = \dfrac{\text{mass of water lost}}{\text{mass of hydrate}} \times 100\%$

From the mass of water lost, moles of water can be calculated. From the mass of the anhydrous salt and the given molar mass of the anhydrous salt.

- molecules of water per formula unit of hydrate (X) = $\dfrac{\text{mole of water}}{\text{mole of anhydrous salt}}$

Example (1):

$$Na_2CO_3.10H_2O(s) \longrightarrow Na_2CO_3(s) + 10\ H_2O(g)$$

In sodium carbonate decahydrate, 10 moles of water molecules are bound to each mole of Na_2CO_3 or 180.20 g of H_2O per 105.99 g of Na_2CO3.

The percent H2O in the hydrated salt is $= \dfrac{180.20\ g}{(180.20\ g + 105.99\ g)} \times 100\% = 62.97\%\ H_2O$

Alums

An alum is a type of chemical compound, usually a hydrated double sulfate salt with the general formula $M^{+1}M^{+3}(SO_4)_2 \cdot XH_2O$, where M^{+1} is a monovalent cation such as potassium or sodium and M^{+3} is a tri-valent ion such as Al^{+3}, Cr^{+3}, often refers to potassium alum, with the formula $KAl(SO_4)_2 \cdot 12H_2O$.

In today's experiment, you will determine the percent by weight of water in a hydrated salt and its formula.

Chemicals and equipment:

- Crucible and cover	- Clay triangle	- Balance	- Tongs
- Desiccator	- Tripod stand	- Bunsen burner	- Hydrated salt

Procedure

1. Weigh your empty clean and dry crucible (without cover).
2. Introduce about 2 g of hydrated salt. Note the appearance and color of the solid; weigh the crucible and contents.
3. Place the crucible on clay triangle that supported by ring over the Bunsen burner as show in Figure (10) Adjust the cover such that the portion of the crucible is uncovered.
4. Start heating the crucible to redness (for about 5 min) using the non-luminous flame, after the salt has shown the indication of water remove.
5. Allow the set to cool in desiccator (it keeps dry while cooling) see Figure (11).
6. Weigh the crucible and content after heat and cool.
7. Heat the crucible and the content for another 2 min then allow cooling as done in previous step, then weighing again.
8. The values between the last two weighing should not differ by more than 0.05g, if the difference is greater than 0.05g repeat step 7 until you achieve constant mass.

Note: At all the time during the work, handle the crucible with tong only.

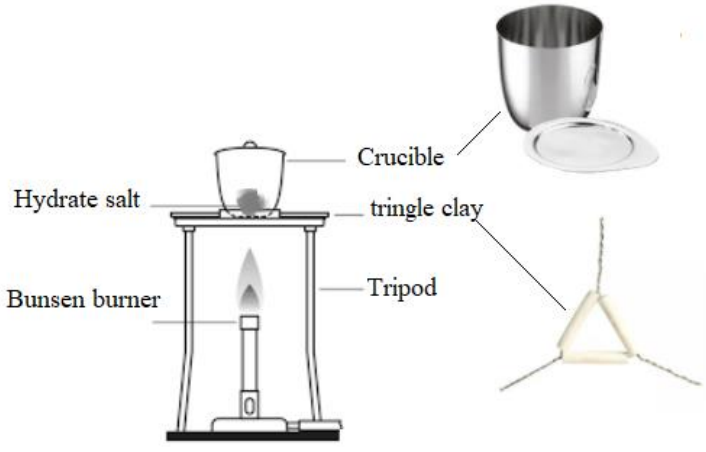


Figure (10)



Figure (11)

Data and calculation:

Unknown No:	***Experiment***	***Duplicate***
(a) Mass of empty crucible	*g*	*G*
(b) Mass of crucible and hydrate	*g*	*G*
(c) Mass of crucible and anhydrous	*g*	*G*
(d) Mass of hydrate [(b) – (a)]	*g*	*G*
(e) Mass of anhydrous [(c) – (a)]	*g*	*G*
(f) Mass of water[(d) – (e)]	*g*	*G*
(g) Percentage of water in hydrate [(f) ÷ (d)] × 100	*%*	*%*
(h) Moles of water [(f) ÷ molar mass (H_2O) =18 g/mol]	*mole*	*Mole*
(i) Molar mass of anhydrous (Mm)	*g/mole*	*g/mole*
(j) Moles of anhydrous [(e) ÷ Mm of anhydrous]	*mole*	*Mole*
(j) Value of X [(h)÷(j)]		
(k) Formula of hydrate		

Show Calculations

……

……

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Questions:

Q1- A hydrated $MgSO_4.XH_2O$ salt, weighing 3.211 g, was heated in a crucible until reaching a constant weight. The weight of the anhydrous $MgSO_4$ was 1.570 grams.

a) Calculate the mass percent of water in the hydrated $MgSO_4$ salt.

b) Calculate the moles of H_2O removed and the moles of anhydrous $MgSO_4$ remaining in the crucible.

c) What is the formula for the hydrated $MgSO_4$?

Q2- In today's experiment what error in the data is likely to occur if the hydrated salt is heated too strongly?

Q3- How long should the hydrated salt be heated in removing the water, and what happened on the value of X if incomplete dehydration is occur?

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 5

Molecular Weight of Volatile Liquids

(Dumas Method)

Student Name:

Student Number:

Submission Date:

Experimental No. 4:

Molecular weight of volatile liquid

(Dumas's method)

Objective

-To determine the molecular weight of an unknown volatile liquid by applying the ideal gas law.

Introduction

Dumas's method was designed by a French chemist named Jean Baptiste André Dumas, after whom the procedure is now named. Dumas used the method to determine the vapors densities of elements (mercury, phosphorus, sulfur) and inorganic compounds.

Theory

One of the important applications of the Ideal Gas Law is found in the experimental determination of the molecular weight of gases and vapors. In order to measure the molecular weight of a gas or vapor, we need simply to determine the mass of a given sample of the gas under known conditions of temperature and pressure. If the gas obeys the Ideal Gas Law,

$$PV = nRT$$

P: pressure in atm (1atm =760 mmHg= 101.3 Kpa =760 torr)

V: gas volume in L (1L=1000mL)

n: the number of moles of the gas = mass/Molar mass (MW)

R: gas constant = 0.08206 L.atm/K.mol

T: temperature in Kelvin (K= °C +273)

This experiment involves measuring the molar mass of a volatile liquid by using the above equation. A small amount of the liquid is introduced into a weighed flask. The flask is then placed in boiling water, where the liquid will vaporize completely, driving out the air and filling the flask with vapor at atmospheric pressure and the temperature of the boiling water. If we cool the flask so that the vapor condenses, we can measure the weight of the vapor and calculate a value for MWt.

Chemicals and equipment:

-Erlenmeyer flask	-Graduated cylinder	-Thermometer (-10 to 110 °C)
-Beaker + water	-Balance	-Volatile liquid
-Aluminum foil	-Rubber band	-Bunsen burner

Procedure

1. Weight empty dry Erlenmeyer flask, aluminum foil and a rubber band on the balance as (m1).
2. Pour about 5 mL of the unknown liquid into the flask. Assemble the apparatus as shown in figure (12).
3. Add a few boiling chips to the water in the 1000 mL beaker and heat the water to boiling. After all the liquid has disappeared and no more vapor comes out of the cap, shut off the burner and measure the temperature.
4. Dry the flask with a towel to remove the surface water. and reweigh the flask as (m2). The difference between m2 & m1 is the mass of condensed vapor.
5. Read the atmospheric pressure from the barometer.
6. Measure the volume of the flask using graduated cylinder after filling the flask with water.

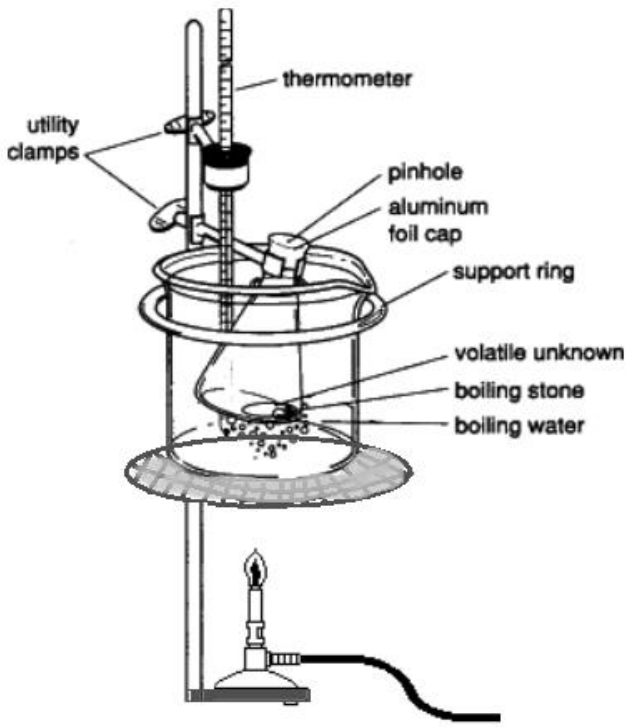


Figure (12)

Data and Calculations:

Mass of empty flask + aluminum foil + rubber band (m_1)	
Mass of empty flask + aluminum foil + rubber band + condensed vapor (m_2)	
Mass of condensed vapor (m_{gas})	
T (°C)	
T (K)	
Pressure (mmHg)	
Pressure (atm)	
Volume of flask (mL)	
Volume of flask (L)	
Molecular weight of volatile liquid (g/mol)	

Show calculations

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Questions:

Q1- A sample of an unknown liquid is vaporized in a flask having a volume of 265 mL at 99.9 °C. The vapor has a mass = 0.725 g, and exerts a pressure = 745 torr. Calculate the gram molecular weight of the unknown liquid.

Q2- Suggest 3 possible sources of experimental error in this experiment.

Q3- A student performing this experiment accidently tore the foil cap, so that extra vapor escaped from the flask before it was removed from the heat. Describe how this would affect the mass of vapor and the calculated molar mass.

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 6

Empirical Formula

Student Name:

Student Number:

Submission Date:

Experiment no. 6:

Empirical Formula

Objective

* The student will be able to determine empirical formula of magnesium oxide experimentally.

Introduction

Every chemical compound has a chemical formula. In fact, there are several different compounds sharing the same chemical formula (molecular formula). Molecular formula is the actual molar ratio between atoms forming certain compound. Moreover, many compounds can have the same empirical formula which represents the simplest molar ratio between atoms of the compound.
Perhaps the most familiar type is the molecular formula – such as H_2O for water and CO_2 for carbon dioxide – which tells us the number of different atoms in each molecule. Structural formulae go a step further by showing how the atoms are linked together within the molecule, which is especially important for organic compounds. Figure 1, for example, shows the empirical, molecular and structural formula for glucose.

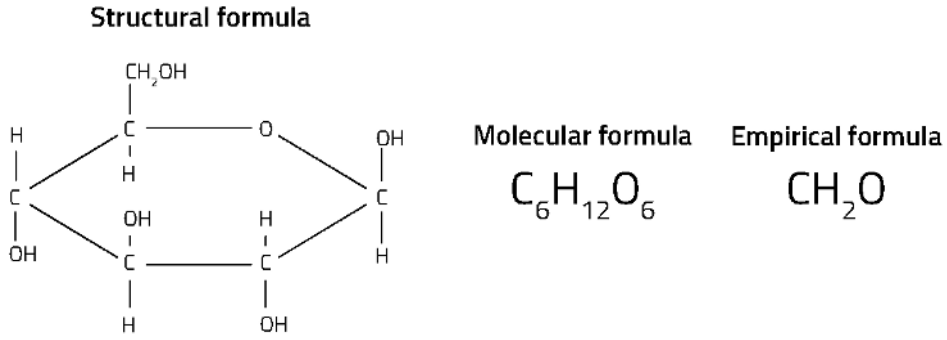


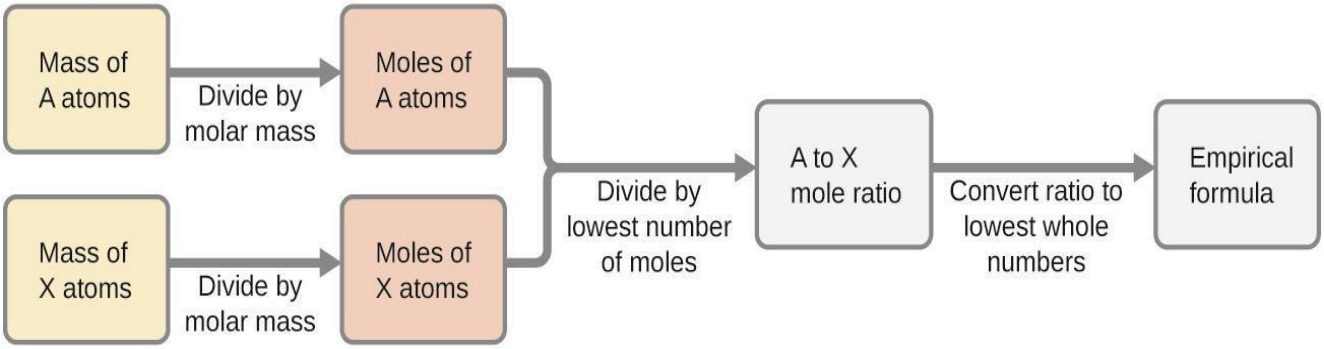
Figure 13: Structural, molecular and empirical formulae for glucose

Theory

The first step for identification a new compound is by determination the empirical formula for it, the empirical formula can be determined experimentally by two methods:

1) The analysis (decomposition) method. 2) The synthetic method.
In each method, empirical formula is derived from experimentally measured element masses

As shown in flow chart:



Example:

If 17.471 g of trioxane is burned, and produced 10.477 g H_2O and 25.612 g CO_2 , what is the empirical formula of this compound?

Trioxane (C,H,O) ⟶ CO_2 + H_2O

Solution:

Because we assume that all the carbon in trioxane has reacted to form in CO_2, we can find the mass of carbon

$$? \text{ g C} = 25.612 \text{ g } \cancel{CO_2} \left(\frac{1 \text{ } \cancel{\text{mol } CO_2}}{44.010 \text{ g } \cancel{CO_2}} \right) \left(\frac{1 \text{ } \cancel{\text{mol C}}}{1 \text{ } \cancel{\text{mol } CO_2}} \right) \left(\frac{12.011 \text{ g } \cancel{C}}{1 \text{ } \cancel{\text{mol C}}} \right) = 6.9899 \text{ g C}$$

Because we assume that all of the hydrogen in trioxane has reacted to form H_2O, we can find the mass of hydrogen

$$? \text{ g H} = 10.477 \text{ g } \cancel{H_2O} \left(\frac{1 \text{ } \cancel{\text{mol } H_2O}}{18.0153 \text{ g } \cancel{H_2O}} \right) \left(\frac{2 \text{ } \cancel{\text{mol H}}}{1 \text{ } \cancel{\text{mol } H_2O}} \right) \left(\frac{1.00797 \text{ g } \cancel{H}}{1 \text{ } \cancel{\text{mol H}}} \right) = 1.1724 \text{ g H}$$

Because trioxane contains only carbon, hydrogen, and oxygen, we can calculate the mass of oxygen by subtracting the masses of carbon and hydrogen from the total mass of trioxane.

$$? \text{ g O} = 17.471 \text{ g trioxane} - 6.9899 \text{ g C} - 1.1724 \text{ g H} = 9.309 \text{ g O}$$

We now calculate the empirical formula, and the molecular formula.

$$? \text{ mol C} = 6.9899 \text{ g C} \left(\frac{1 \text{ mol C}}{12.011 \text{ g C}} \right) = 0.58196 \text{ mol C} \div 0.5818 \approx 1 \text{ mol C}$$

$$? \text{ mol H} = 1.1724 \text{ g H} \left(\frac{1 \text{ mol H}}{1.00794 \text{ g H}} \right) = 1.1631 \text{ mol H} \div 0.5818 \approx 2 \text{ mol H}$$

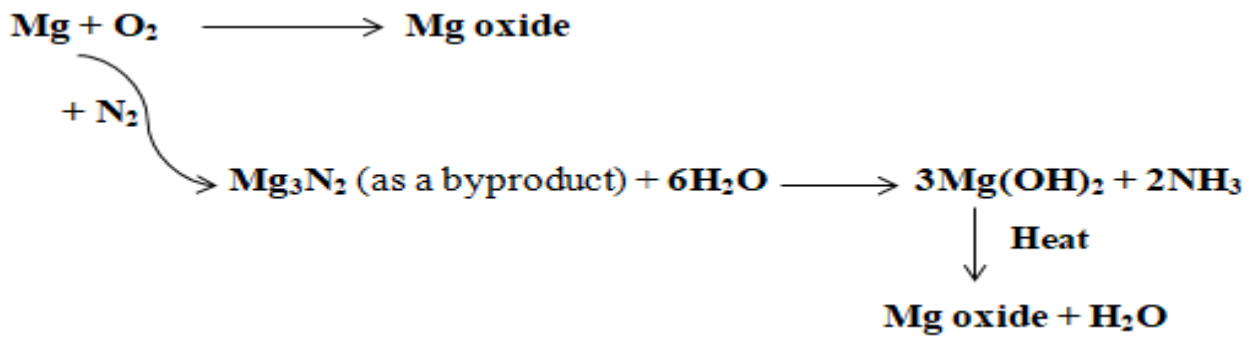
$$? \text{ mol O} = 9.309 \text{ g O} \left(\frac{1 \text{ mol O}}{15.9994 \text{ g O}} \right) = 0.5818 \text{ mol O} \div 0.5818 = 1 \text{ mol O}$$

* Empirical formula: CH_2O

* $$n = \frac{\text{molecular mass}}{\text{empirical formula mass}} = \frac{90.079}{30.026} = 3$$

* Molecular formula: $C_3H_6O_3$

In this experiment you will determine the empirical formula for an ionic compound. This experiment can be performed by weighing certain amount of magnesium tunings and allowing it to be burned in air to produce magnesium oxide using the synthetic method in a covered crucible. Because air contains both oxygen and nitrogen gases, magnesium will react with both of them, but with oxygen faster than nitrogen, as shown below:



chemicals and equipment

- Crucible and cover.	- Magnesium turnings.	- Iron ring.	
-Clay tringle.	- Stands and clamps.	- Balance	
- Tongs	- Droppers	- Distilled water.	– PH paper

Procedure

1) Obtain crucible and its cover from your instructor.

Note: *(check the cleaned of your crucible if it isn't clean, clean your crucible as your instructor orders).*

2) Weigh the empty crucible without cover (m1).

Note: (*at this stage crucible should be handled with tongs only).*

3) Put about 0.25g of magnesium tuning in the crucible and weigh again (m2).

4) Construct the set up in (figure 14) adjusts the cover to be slightly open to allow air to enter the crucible.

5) Heat the crucible and its content until all magnesium convert to white or slightly grey (ash) and no glow occurs.

Note: (*Don't open the crucible widely; otherwise, magnesium will burn brightly, if this happen covers the crucible immediately).*

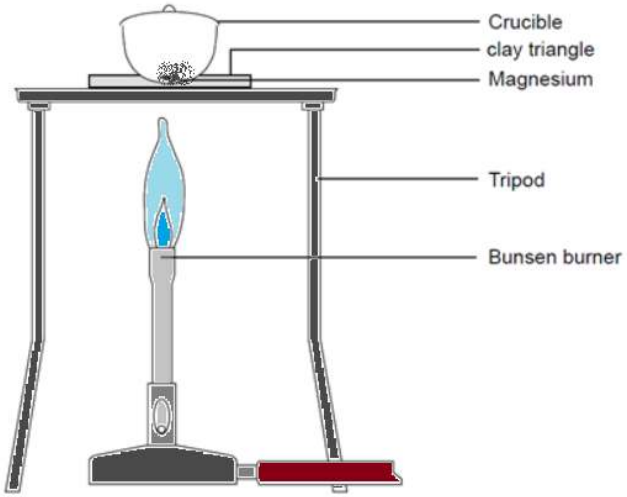


Figure (14)

6) Allow the crucible and its contents to cool. Add about 1 ml (~10 drops) of deionized water directly to the solid powder. Carefully waft some of the gas that is generated toward your nose, but ***be very careful***. Record any odor. Also, use tongs to carefully hold a piece of moist red litmus/PH paper over the crucible. Record your observations.

7) Reheat the crucible and its content to drive of the water that has been added for (5-8 min).

8) Allow the crucible to cool completely at room temperature obtain the mass of crucible and record it.

9) Reheat the crucible and contents for another two min then allow cooling as done in the previous step, then weigh again. The values obtained in steps 8 and 9 should not differ by more than 0.05g, if the different is more than 0.05g, repeat step (9) until you achieve constant mass.

10) Ask your instructor about discarding the magnesium oxide and clean your crucible carefully.

Data and Calculation

		Unit
Mass of empty crucible (m_1)		
Mass of crucible + Magnesium (m_2)		
Mass of crucible + Magnesium Oxide (m_3)		
Mass of Magnesium Oxide $m_{Mg_X O_Y}$ (m_3-m_1)		
Mass of Magnesium m_{Mg} (m_2 - m_1)		
Moles of Magnesium X =m_{Mg} \24.3 (g\mol)		
Mass of Oxygen $m_O = m_{(Mg\ X\ OY)} - m_{Mg}$		
Moles of Oxygen Y =m_O \16(g\mol)		
The mole ratio of Magnesium to Oxygen X:Y		
The Empirical formula of Magnesium oxide		

<u>Show calculation:</u>

1- Moles of magnesium =

2- Moles of oxygen =

3- The Mole ratio =

Questions:

1) How would the calculated molar ratio between (Mg/O) be affect if:

a- Forget the addition of water to get rid of magnesium nitride form.

b- Incomplete dehydration was done after water was added.

c- Some of magnesium oxide spilled out before final weighting.

1) Calculate Mg% in your compound in this experiment.

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 7

Precipitation Reaction (Limiting Reactant)

Student Name:

Student Number:

Submission Date:

Experiment No.7:

Precipitation Reaction (Limiting Reactant)

Objectives

1-The student will be able to determine the limiting reactant in the reaction of sodium carbonate and calcium chloride from the moles of reactant.

2- The student will be able to calculate the percentage yield of the precipitate.

Introduction

Stoichiometry is a process of "keeping track" of the quantities of reactants and products in a chemical reaction. Sometimes in a chemical reaction, the reactants may not be mixed in the same proportions. As a result, there is a limiting reagent and an excess reagent. A limiting reagent is the reactant that limits the amount of product formed in a chemical reaction. An excess reagent is the reactant that not completely used up in a chemical reaction, as the following example:

Consider the process it takes to make a sandwich. You need two slices of bread and one piece of beef to make each sandwich. How many complete sandwiches could you make if you had eighteen slices of bread and six slices of beef? Let's set this up like a chemical equation where the ham and bread are our reactants and the sandwich is our product:

Two slices of bread + 1 piece of beef = 1 burger sandwich



Figure 15: *A practical example clarifying the meaning of limiting reactant.*

Now we can use stoichiometry to determine the number of sandwiches we could make if we used all of our reactants.

Six slices of beef make six sandwiches and each sandwich need two slices of bread. So, we use 12 slices of bread. Since the beef limits the number of sandwiches we can make, the ham is our limiting reagent and the bread is going to be in excess when the beef is consumed by the reaction

Theory

The limiting reactant can be defined as the reactant which consumed first and on which the amount of products depends on.

In this experiment, you will predict and observe a limiting reactant during the reaction which involves a precipitation of $CaCO_3$ by mixing definite volumes solutions of known molarity of calcium chloride and sodium carbonate as the following equation:

$$CaCl_{2\,(aq)} + Na_2CO_{3\,(aq)} \rightarrow CaCO_{3(s)} + 2NaCl_{\,(aq)}$$

Once the limiting reactant is known, the theoretical and percentage yields can be calculated.

* Determine the limiting reagent if 76.4 grams of $C_2H_3Br_3$ reacts with 49.1 grams of O_2.

$$4\ C_2H_3Br_3 + 11\ O_2 \longrightarrow 8\ CO_2 + 6HO_2 + 6Br_2$$

Solution:

Using method 1,

$$76.4\ g \times \frac{1\ mol}{266.72g} = 0.286 \text{ moles of } C_2H_3Br_3$$

$$49.1\ g \times \frac{1\ mol}{32g} = 1.53 \text{ moles of } O_2$$

If you assume that all of the oxygen is used up,
(1.53 x 411), or, 0.556 moles of $C_2H_3Br_3$ are required. Since there are only 0.286 moles of $C_2H_3Br_3$ that are available, $C_2H_3Br_3$ is the limiting reagent here.

Using method 2,

$$76.4\ g\ C_2H_3Br_3 \times \frac{1\ mol}{266.72g} \times \frac{8\ mol\ CO_2}{4\ mol\ C_2H_3Br_3} \times \frac{44.01\ g\ CO_2}{1\ mol\ CO_2} = 25.2\ g\ CO_2$$

$$49.1\ g\ O_2 \times \frac{1\ mol\ O_2}{32\ g\ O_2} \times \frac{8\ mol\ CO_2}{11\ mol\ O_2} \times \frac{44.01\ g\ CO_2}{1\ mol\ CO_2} = 49.1\ g\ CO_2$$

Hence, by using any of these methods, $C_2H_3Br_3$ is the limiting reagent.

chemicals and equipment

- Two Beakers 50 ml.	- Glass rod.	- Buchner Funnels.
- Graduated cylinder 25 ml.	- Filter papers.	- Oven
- $CaCl_2$ solution	- $NaCO_3$ solution	

Procedure

1) Into clean beakers take 15 ml volume of $CaCl_2$ and Na_2CO_3 solutions (with definite molarity given by your instructor).

2) Into a clean 100 ml Erlenmeyer flask mix the two solutions together.

3) Swirl the flask gently and observe the formation of the precipitate.

Note: If the precipitate does not form, scratch the walls of flask with glass rode.

4) Filter the solid produced (as shown in figure 16), wash it with cold water, and allow it to dry in oven.

5) Weigh the solid collected and determine the percentage yield of the reaction.

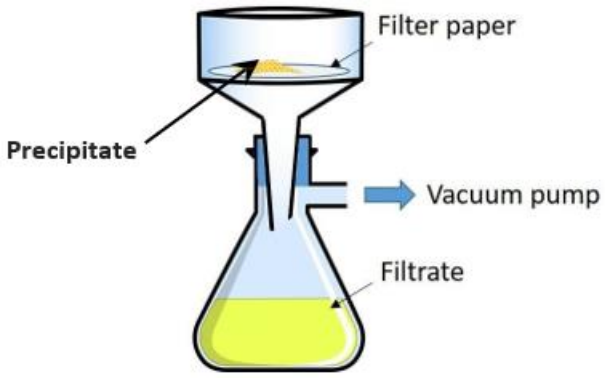


Figure 16

Useful Data	
Substance	Molar mass (g/mol)
$CaCl_2$	110.98
Na_2CO_3	105.99
$CaCO_3$	100.09

Data and calculations:

** Determination of the limiting reactant:*

Data		**Unit**
Volume of $CaCl_2$		
Molarity of $CaCl_2$		
Volume of Na_2CO_3		
Molarity of Na_2CO_3		
Number of moles of $CaCl_2$ reacted (M x V_L)		
Number of moles of Na_2CO_3 reacted (M x V_L)		
The Limiting reactant is		
The Excess reactant is		
The mass of $CaCO_3$ expected (Theoretical yield)		
The mass of $CaCO_3$ precipitate (Actual yield)		
% Yield of $CaCO_3$ = $\frac{Actual\ Yield}{Theoretical\ Yield} \times 100\%$		

Show calculations

Moles of $CaCl_2$	*Moles of Na_2CO_3*
Moles of $CaCO_3$ expected	*% Yield of $CaCO_3$*

Questions:

1) If 23 grams of iron (II) chloride ($FeCl_2$) reacts with 41 grams of sodium phosphate (Na_3PO_4), what is the limiting reagent? How much sodium chloride can be formed?

$$3FeCl_2 + 2Na_3PO_4 \longrightarrow Fe_3(PO_4)_2 + 6NaCl$$

2) How much of the excess reagent remains when this reaction has gone to completion?

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 8

Volumetric Analysis I (Vinegar Analysis)

Student Name:

Student Number:

Submission Date:

Experiment No. 8:

Volumetric Analysis (I)/Vinegar Analysis

Objectives

1) To understand the meaning of volumetric analysis.
2) Use pipet and burettes correctly.
3) Perform a titration precisely.
4) To determine the Molarity and mass percent (m/m %) of acetic acid (CH_3COOH) in vinegar, by acid base titration.

Introduction

A technique known as **titration** will be used in this experiment to determine the concentration of acetic acid in vinegar. A titration involves performing a reaction between a solution of known concentration (**the titrant**) and a solution of unknown concentration (**the analyte**). In this experiment, the titrant is sodium hydroxide (NaOH) solution (a basic solution) and the analyte is acetic acid (CH_3COOH) in vinegar (acidic solution). When mixed together, a **neutralization reaction** occurs between sodium hydroxide and the acetic acid in vinegar. A pH indicator is used to find out the end point of titration.

Theory

Vinegar is essentially a solution of acetic acid (CH_3COOH) in water. The concentration of acetic acid in vinegar may be expressed as a molarity (mol/L)

$$\textbf{Molarity} = \frac{\textbf{Moles of } CH_3COOH}{\textbf{Volume of vinegar solution (in L)}}$$

or mass percent:

$$\%\ \text{acetic acid in vinegar solution} = \frac{\text{grams of acetic acid}}{\text{grams of vinegar solution}} \times 100\,\%$$

In this experiment, vinegar is analyzed (for acetic acid) by an **acid- base titration** with NaOH. The chemical equation shows the reaction between NaOH and vinegar:

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

According to the balanced chemical equation shown above, at equivalence point (end-point);

moles of NaOH = moles of acetic acid

The sodium hydroxide will be gradually added to the vinegar in small amounts from a **burette**. A burette is a graduated glassware with a tap at one end that allows the precise delivery of known volumes of a solution (the titrant).
pH indicators are chemical substances that are able to change their color depending on pH of the solution, they are used in acid- base titrations, to detect the **equivalence point** of titration, i.e., the point at which the reaction between acid and base is completed. For example, phenolphthalein, the indicator used in this experiment, is colorless in the acidic medium, but turns into pale pink color in the basic solution. This point at which one drop of titrant (NaOH solution) changes the color of solution is called the **end point.**

Actually, the equivalence point and the end point *are not exactly the same*, the equivalence point is a theoretical point, determined by stoichiometry of the reaction, whereas the end point is an experimental (observable) point, it is the point at which the indicator's color changes.

***Safety**:* Be careful when handling the sodium hydroxide base (NaOH), as it is corrosive and can cause chemical burns to the skin. If any NaOH spills on you, rinse immediately under running water for up to 15 minutes and report the accident to your instructor.

Chemicals and equipment:

- 250-mL Erlenmeyer flask	- dropper	- 10mL cylinder	- burette stand
– funnel	- wash bottle with distilled water		- 50mL burette
- 10mL volumetric pipette	- pipette filler		- vinegar solution
- 0.1M NaOH standard solution	- Phenolphthalein indicator		

Procedure

1. Rinse the inside of the burette with distilled water. Allow the distilled water to drain out through the tip (stopcock) in order to ensure that the tip is also rinsed.

2. Rinse the burette with a small amount of NaOH solution, and swirl the burette on its side (over the sink) to rinse its entire inner surface. Then allow the NaOH solution to drain out through the tip.

3. Fill the burette with NaOH solution up to the top. Use a funnel to do this carefully, preferably over the sink. Now measure the volume at the level of the NaOH precisely, and record it as the "**Initial Burette Reading"** on your report. Also record the exact molarity of the NaOH, which is labeled on the stock bottle.

4. Use the volumetric pipette to transfer 10.00-mL of vinegar solution into a clean 250 mL Erlenmeyer flask. Then add 5 drops of phenolphthalein to this Erlenmeyer flask.

5. Begin the titration by slowly adding NaOH solution from the burette to the vinegar in the Erlenmeyer flask. Swirl Erlenmeyer flask as you add the base in order to efficiently mix the chemicals. Some pinkness may appear briefly in the flask as the base is added, but it will quickly disappear as the flask is swirled.

6. As the equivalence point is approached, the pink color will become more pervasive and will take longer time to disappear. When this occurs, start
to add the NaOH **drop by drop**. Finally, the addition of just one drop of NaOH solution will turn the solution in the Erlenmeyer flask to a pale pink color that does not disappear when swirled. This indicates that the equivalence point has been reached. **Do not add any more** NaOH **at this point.**

7. Measure this volume of NaOH precisely, and record it as the "**Final Burette Reading**" on your report. Then show the resulting solution in the flask to your instructor so s/he can record the final color on your report form.

8. When finished, dispose of your chemical waste as instructed.

Helpful information:

Molecular weight of Acetic acid = 60 g/ mol

Density of Vinegar = 1.07 g/mL

Data and Calculations

		Unit
Volume of diluted vinegar solution		
Molarity of NaOH solution		
Initial burette reading of NaOH solution		
Final burette reading of NaOH solution		
Volume of NaOH solution		
Moles of NaOH		
Moles of CH_3COOH		
Mass of CH_3COOH		
Molarity of diluted vinegar solution		
Mass percentage of CH_3COOH in solution ($\%CH_3COOH$)		

Show calculations:

QUESTIONS

1. What was the purpose of the phenolphthalein indicator in this experiment?

2. Suppose you added 40 mL of water to your vinegar sample instead of 20 mL. Would the titration have required more, less or the same amount of NaOH solution for a complete reaction? Explain.

3. Suppose you had titrated your vinegar sample with barium hydroxide instead of sodium hydroxide:

   $\mathbf{Ba(OH)_{2(aq)} + 2CH_3COOH_{(aq)} \rightarrow Ba(CH_3COO)_{2(aq)} + 2H_2O_{(l)}}$
   Consider a 0.586 M aqueous solution of barium hydroxide,
   What volume (in mL) of 0.586 M $Ba(OH)_2$ solution are required to neutralize 10 ml of vinegar containing 2.78 g of acetic acid?

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 9

Volumetric Analysis II (Bleach Analysis)

Student Name:

Student Number:

Submission Date:

Experiment No. 9:

Volumetric Analysis II/ Bleach Analysis

Objectives

1) In this experiment, the molarity of sodium hypochlorite in ***Clorox bleach*** is determined through indirect titration (back titration) with a standard solution of sodium thiosulfate solution.
2) To understand the difference between direct titration and back titration

Introduction

Nature and Production of Bleach.

Liquid household bleaches usually contain approximately 3-8% sodium hypochlorite (NaOCl) by weight; it is the oxidizing power of the hypochlorite ion that is responsible for the beneficial action of bleach. Hypochlorite performs its bleaching function by oxidizing stains (or dyes) to produce colorless, soluble, or gaseous species. Many at laundry novices discover the hard way that bleach splashed on clothes leads to white spots.

Liquid laundry bleach is prepared commercially by electrolysis of a cold, stirred solution of sodium chloride.

$$2\ Cl^-_{(aq)} \rightarrow Cl_{2(g)} + 2e^- \qquad \textit{Oxidation}$$

and hydroxide ion is formed at the cathode:

$$2\ H_2O_{(l)} + 2e^- \rightarrow 2\ OH^-_{(aq)} + H_{2(g)} \qquad \textit{Reduction}$$

In solution, the chlorine gas and hydroxide ion react to give **hypochlorite ion**:

$$Cl_{2(g)} + 2\ OH^-_{(aq)} \rightarrow \mathbf{OCl^-}_{(aq)} + H_2O_{(l)}$$

In the aqueous solution, sodium hypochlorite dissociates into Na^+ and OCl^-. The hypochlorite ion is determined by performing a reduction-oxidation titration (redox titration) between the oxidizing agent (NaOCl) in bleach, and the reducing agent (KI).

Theory

The diluted bleach containing NaOCl is mixed with a solution containing excess I^- ion, a reducing agent, which reduces the hypochlorite ion to a chloride ion (Cl^-). The iodide ion (I^-) is oxidized to iodine (I_2).

$$OCl^-_{(aq)} + 2I^-_{(aq)} + 2H_3O^+_{(aq)} \rightarrow I_{2(aq)} + Cl^-_{(aq)} + 3H_2O_{(l)}$$

It is possible to measure the amount of iodine produced by this reaction through a redox titration with a solution of thiosulfate ion ($S_2O_3^{2-}$).

$$I_{2\,(aq)} + 2S_2O_3^{2-}{}_{(aq)} \rightarrow 2I^-_{(aq)} + S_4O_6^{2-}{}_{(aq)}$$

brown colorless colorless colorless

A color change occurs when the thiosulfate combines with the iodine solution as the solution begins a deep brown color, and then it becomes colorless as the reaction products are formed. Starch can be added (as an indicator) to see this color change more clearly.

Once the starch is added to the iodine solution, a dark blue color appears, indicating the formation of the iodine-starch complex. The iodide solution is back titrated with the $Na_2S_2O_3$ solution until the blue color disappears.

From the above chemical equations, notice that:

$$\text{Moles of } I_2 = \tfrac{1}{2} \text{ Moles of } S_2O_3^{2-}$$

$$\text{Moles of } OCl^- = \text{Moles of } I_2$$

The number of moles of **thiosulfate** can be determined through the titration. This leads to the number of moles of **iodine** which in turns allows the determination of the moles of **hypochlorite** in the diluted bleach. Finally, the number of **hypochlorite** moles in the original bleach solution can be calculated.

Chemicals and equipment:

- 250 mL Erlenmeyer flask -50 mL Burette - Dropper - funnel
- 10 mL cylinder - washing bottle - 5.00 mL of commercial bleach
- 2 mL of 2 M HCl solution. - 0.1M $Na_2S_2O_3$ standard solution - Starch
- 2.0 grams of solid KI - Distilled water

Procedure

1) Transfer 5.00 mL of commercial bleach solution into a 100 mL volumetric flask.
2) Dilute the solution with exact amount of distilled water and mix well.
3) Add roughly 2.0 grams of solid KI to the flask.
4) Add 25 mL of distilled water to the volumetric flask.
5) Swirl the flask to dissolve the KI, and then add 2 mL of 2 M HCl. A brown color should appear in this step, once seen, place the flask containing the brown iodine solution in a dark place for minimum 10 minutes, *to avoid photooxidation of iodide ion (I^-).*
6) A cleaned-burette is set up and filled up with 0.100 M sodium thiosulfate solution.
7) The initial burette volume reading is recorded.
8) The solution is then titrated until the iodine color of red-brown fades to light brown- yellow color.
9) Then, a dropper full of starch solution is added, which should turn the solution to a dark blue color. But, if not, you should repeat your work again.
10) Continue the titration until the blue color disappears. (As shown in figure 17)

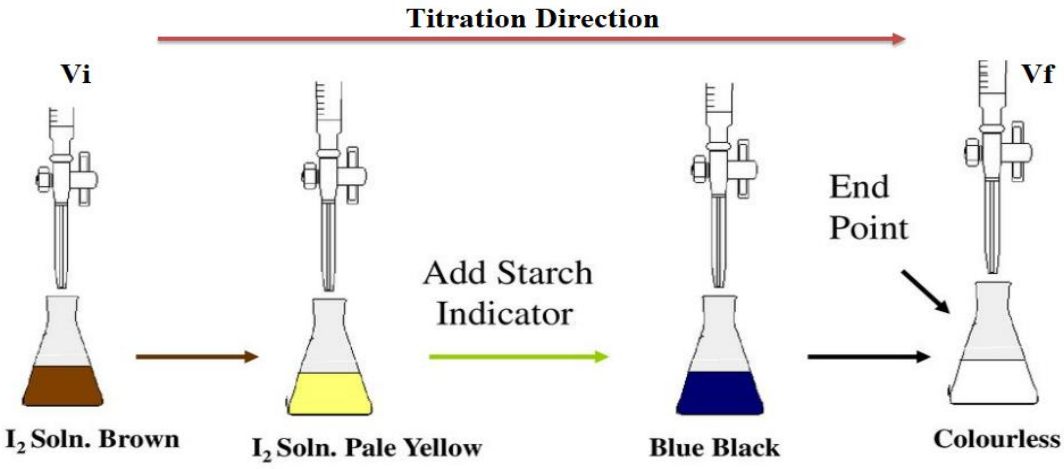


Figure (17)

11) Record the final burette volume reading in the data table.

Repeat the titration procedure for more results.

Helpful information:

Bleach Density = 1.02 g/mL

NaOCl Molecular weight = 74.5 g/mol

Cl_2 Molecular weight = 71 g/mol

Data and calculation:

	Trial 1	Trial 2	Unit
Volume of diluted bleach solution			**mL**
Dilution factor			
Molarity of $S_2O_3^{-2}$ solution			**M**
Initial burette reading of $S_2O_3^{-2}$ solution			**mL**
Final burette reading of $S_2O_3^{-2}$ solution			**mL**
Volume of $S_2O_3^{-2}$ solution			
Moles of $S_2O_3^{-2}$			
Moles of I_2			
Moles of OCl^-			
Molarity of diluted bleach			
Molarity of original concentrated bleach			

* Mass percentage of NaOCl in original bleach solution (%NaOCl) (show your calculations)

**Mass percentage of Cl_2 in original bleach solution (%Cl_2) (show your calculations)

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 10

Thermochemistry and Calorimetry

Student Name:

Student Number:

Submission Date:

Experiment No. 10:

Thermochemistry and Calorimetry

Objectives

1) To measure the enthalpy of the reaction between Mg and HCl.
2) To measure the enthalpy of the reaction between MgO and HCl.
3) To calculate the enthalpy of formation of MgO.

Introduction

Thermodynamics is the part of science dealing with the heat change accompanied to a chemical or physical processes. Thermochemistry as a branch of thermodynamics representing the quantitative measurements of the heat gained or evolved during chemical processes.

The enthalpy change can be measured and calculated directly as by Calorimetry or indirectly. Indirect methods including; Hess's Law, the enthalpy of formation and the bond energy differences between products and reactants.

The enthalpy of a process is the heat content under a constant pressure. It can be easily measured via cup Calorimetry, when the process is a chemical reaction carried out at constant pressure. It can also be predicted from the difference between the enthalpies of the products and reactants.

However, if the reaction involves gases, cup-Calorimetry will be not suitable. In this case the proper method is called bomb-Calorimetry and the instrument used is the bomb-calorimeter.

Theory

Cup calorimeter

Although Calorimetry is simple in principle, its practice is a highly exacting art, especially when applied to processes that take place slowly or involve very small heat changes, such as the germination of seeds. Calorimeters can be as simple as a foam plastic coffee cup, which is often used in student laboratories. Research-grade calorimeters, able to detect minute temperature changes, are more likely to occupy table tops, or even entire rooms.

A coffee - cup calorimeter is a constant pressure calorimeter. As such, the heat that is measured in such a device is equivalent to the change in enthalpy. A cup calorimeter is typically used for solution based chemistry and as such generally involves a reaction with little or no volume change. Therefore, while the heat is exactly equal to the enthalpy change (ΔH) in this case, it will also be approximately equal to the change in the internal energy (ΔU) since the work will be very small (assuming there are no gas reactants or products).
The more technical name for this type of Calorimetry is isobaric Calorimetry. The "coffee-cup" name comes from the fact that most of the time this experiment is done inside of a simple

Styrofoam cup. A Styrofoam cup makes for a good adiabatic wall and helps keep all the heat released or absorbed by the reaction inside the cup so we can measure it.

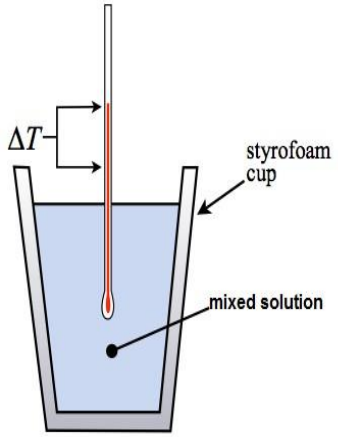
The heat capacity of a coffee cup calorimeter is typically taken to be that of the water in the calorimeter. However, it could be that the "hardware" and the water heat capacities have both been measured and you treat it much like in bomb calorimetry.

To calculate the value of enthalpy of reactions in solutions we use cup- calorimetry in which:

$q_{rxn.}$ (q_{system}) = $- q_{surrounding} = - (q_{solution} + q_{cup})$

Assuming an adiabatic system (the cup walls has no conduce), then $q_{cup} = 0$ and hence, $q_{rxn} = - q_{soln.}$

$q_{solution} = m.S_p.\Delta T$ (q is the heat in J, m is the mass of solution, Sp is the specific heat of the solution and $\Delta T = T_{final} - T_{initial}$)



coffe-cup calorimeter

$q_{surr.} = m_{solution} C_{solution} \Delta T = - q_{system(rxn)}$

$\Delta H_{rxn} = q_{system(rxn)} / n$ *(n = number of moles)*

Note: be careful with the sign on ΔH. Heat *out* of the reaction is the negative of the heat *in* to the calorimeter and vice versa. So, an increase in temperature of the solution means the reaction was "exothermic" and ΔH is negative. Conversely, if the temperature drops during the reaction, then the reaction must be endothermic and ΔH is positive.

Hess's law

Hess's Law states that; regardless of the multiple stages or steps of a reaction, the total enthalpy change for the reaction is the sum of all individual changes. This law emphasis that the enthalpy change is a state function**.**

Examples:

Example 1: Calculating the enthalpy of dissociation of acetic acid:

$CH_3COOH_{(aq)} \rightarrow CH_3COO^-_{(aq)} + H^+$, $\Delta H_{diss} = ???$

Using the following reactions;

1- $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$ $\Delta H_1 = -55.89$ kJ /mol
2- $NaOH_{(aq)} + CH_3COOH_{(aq)} \rightarrow CH_3COONa_{(aq)} + H_2O_{(l)}$ $\Delta H_2 = -56.14$ kJ /mol

By applying Hess's law $\Delta H_{diss} = \Delta H_2 + (- \Delta H_1) = -0.25$ kJ /mol

Example 2: The enthalpy of formation of magnesium oxide is the enthalpy of the following reaction:

$$Mg_{(s)} + \tfrac{1}{2}O_{2\,(g)} \longrightarrow MgO_{(s)} \qquad \Delta H_{form}$$

Directly measuring the enthalpy of this reaction is challenging, however we can determine this enthalpy from the enthalpy of solution of magnesium metal

$$Mg_{(s)} + 2\,H^{+}_{\,(aq)} \longrightarrow Mg^{2+}_{(aq)} + H_{2\,(g)} \qquad \Delta H_1$$

the enthalpy of salvation of magnesium oxide

$$MgO_{(s)} + 2\,H^{+}_{\,(aq)} \longrightarrow Mg^{2+}_{(aq)} + H_2O_{(\ell)} \qquad \Delta H_2$$

and the enthalpy of formation of liquid water

$$H_{2\,(g)} + \tfrac{1}{2}\,O_{2\,(g)} \longrightarrow H_2O_{(\ell)} \qquad \Delta H_3 = -285.8 \text{ kJ/mol.}$$

Equipment and Reagents

1) Styrofoam cups
2) Graduated cylinder
3) Thermometer (-10 to 110 °C)
4) 250 mL Beaker
5) Balance
6) HCl 0.5M
7) Mg powder
8) MgO powder

Procedure

Part 1: (Mg + HCl)

1. Weigh an empty dry-cup and record the weight**.**
2. Add 25 mL of 0.5 M HCl into the weighed cup calorimeter and measure the temperature of the HCl. Record the temperature.
3. Weigh approximately 0.1 Mg. Record the exact mass of the sample.
4. Add the Mg to the calorimeter and swirl to dissolve. Record the temperature.
5. Weigh the cup calorimeter and the solution in it and record the exact mass.

Part 2: (MgO + HCl)

1. Weigh an empty dry-cup and record the weigh
2. Add 25 mL of 0.5 M HCl into the weighed cup calorimeter and measure the temperature of the HCl. Record the temperature
3. Weigh approximately 0.2 MgO. Record the exact mass of the sample.
4. Add the MgO to the calorimeter and swirl to dissolve. Record the temperature.
5. Weigh the cup calorimeter and the solution in it and record the exact mass

Data &calculation

Part 1:

	Value	unit
Mass of empty cup-calorimeter m_1		
Mass of empty cup-calorimeter + $MgCl_2$ solution m_2		
Mass of $MgCl_2$ solution (m_2-m_1)		
T_i		
T_f		
ΔT		
Heat gained by the calorimeter $q_{cal.}$		
Heat gained by the solution $q_{sol.}$		
Heat lost by the reaction $q_{rxn.}$		
Mass of Mg m_{Mg}		
Number of moles of Mg n_{Mg}		
ΔH_1		

Part 2:

	Value	unit
Mass of empty cup-calorimeter (m_1)		
Mass of empty cup-calorimeter + $MgCl_2$ (m_2)		
Mass of $MgCl_2$ solution (m_2-m_1)		
T_i		
T_f		
ΔT		

Heat gained by the calorimeter	q_{cal}		
Heat gained by the solution	q_{sol}		
Heat lost by the reaction	q_{rxn}		
Mass of MgO			
Number of moles of MgO	n_{MgO}		
ΔH^o_2			

Calculation of ΔH_f^o

$\Delta H^o_{form} = \Delta H^o_1 + (-\Delta H^o_2) + \Delta H^o_3$

$\Delta H^o_{form.}$ = ----------

Post lab Questions:

Q1- Use the thermochemical equations given below to calculate the enthalpy change for the following reaction.

$N_2O_{4(g)}$ ------> $2NO_{2(g)}$ $\Delta H = ?$

(1) $N_{2(g)} + 2O_{2(g)}$ ------> $2NO_{2(g)}$ $\Delta H = +66.36$ kJ\mol

(2) $N_{2(g)} + 2O_{2(g)}$ ------> $N_2O_{4(g)}$ $\Delta H = +9.16$ kJ\mol

Q2- A 113.25 gram sample of gold is initially at 100.0 $^{\circ}C$. It gains 20.00 J of heat from its surroundings. What is its final temperature? (specific heat of gold = 0.129 J g^{-1} $^{\circ}C^{-1}$)

Faculty of Engineering Technology

Physics and Basic science Department

General chemistry lab 1

Course Number: 30206111

Experiment No. 11

Chemical Kinetics

Student Name: Student

Number: Submission

Date:

Experiment No. 11

Chemical Kinetics

Objectives

The main object of this experiment is to study the kinetics of the reaction between Iodide Ions and Hydrogen Peroxide. Through this study the student will be able;

- To determine the rate law of a chemical reaction using the Method of Initial Rates.
- To determine the activation energy of the reaction by finding the value of the rate constant, k, at several temperatures.
- To observe the effect of a catalyst on the reaction rate.

Introduction

Chemical kinetics deals with the speed, or rate of a reaction and the mechanism by which the reaction occurs. We can think of the rate as the number of events per unit time. The rate at which you drive (your speed) is the number of miles you drive in an hour (mi/h). For a chemical reaction the rate is the number of moles that react in a second. In practice, we usually monitor how much the concentration (the number of moles in a liter) changes in a second. Reaction rates are usually expressed in units of moles per liter per second, or molarity per second (M/s).

Chemical kinetics in a chemical reaction ~~it~~ is the amount of product that forms in a given interval of time or it can be defined as the amount of reactant that disappears in a given interval of time.

Reaction rates can be affected in a number of ways. The four factors are:

1. Concentration.
2. Nature of the reactants.
3. Temperature.
4. The presence of a catalyst.

Theory

Consider the reaction, R $\longrightarrow$ P

Assuming that the volume of the system remains constant, one mole of the reactant produces one mole of the product. If $[R_1]$ and $[P_1]$ are the concentrations of R and P respectively at time t_1 and $[R_2]$ and $[P_2]$ are the concentrations of R and P respectively at time t_2, then,

The decrease in the concentration of the reactant, $\Delta R = R_2 - R_1$

The increase in the concentration of the product, $\Delta P = p_2 - p_1$ Time

take for the reaction, $\Delta t = t_2 - t_1$

Rate of disappear of reactant = $\frac{\text{Decrease in the concentration of the reactant}}{\text{Time taken}} = \frac{-\Delta R}{\Delta t}$

Rate of appearance of product = $\frac{\text{Increase in the concentration of the product}}{\text{Time taken}} = \frac{\Delta P}{\Delta t}$

Therefore, the rate of the reaction = $\frac{-\Delta R}{\Delta t} = \frac{\Delta P}{\Delta t}$

Reaction rates often depend on reactant concentrations. In a number of simple cases, the rate is proportional to a reactant's concentration, raised to some power.
Which is the **rate law**. The general form of the rate law is

$$\textbf{Rate} = k[A]^a[B]^b$$

where **A** and **B**, are reactants or products. The proportionality constant, **k**, is called the rate constant. The units of the rate constant will depend upon the form of the rate law. The rate constant is generally a function of temperature.

Here, the reaction is m^{th} order in a and n^{th} order in b; its overall order is m + n.

The proportionality constant, k, which is called the reaction rate constant, depends on temperature and is given by the Arrhenius equation.

$$k = Ae^{-Ea/RT}$$

Where *A* is a constant called the *Arrhenius constant,* E_a the *activation energy* for the reaction; *R* is the universal gas constant in units of J/K · mol; and, *T* is the absolute temperature in Kelvin. You will determine the rate law for reaction (m, n, *A*, and E_a). This enables *the rate* for *any concentrations at any temperature* to be calculated.

$$\ln\frac{k_2}{k_1} = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Equipment and Reagents:

- Graduated cylinders, (10, 25, 50) mL	- Conical flask (250 ml)
- Two 50ml burette	- volumetric pipet 10.00 mL
- H_2O_2 (3%)	- $Na_2S_2O_3$ (0.05M)
- Acetate buffer PH 4.7	- KI solution (0.1 M)
- 1% starch solution	-Distill water
- Hot-water bath.	

In today's experiment, the balanced chemical equation of the reaction under study is:

$$H_2O_{2\,(aq)} + 2I^-_{(aq)} + 2H^+_{(aq)} \longrightarrow I_{2(aq)} + 2H_2O_{(l)}$$

The rate of the above reaction will be determined.

$$\text{Rate} = k[H_2O_2]^m[I^-]^n[H^+]^p$$

Rate dependence on concentrations of the reactants:

Let us understand the dependence of the reactant concentration on the rate of chemical reaction by studying the reaction between iodide ions and sodium thiosulphate in the presence of hydrogen peroxide.

Hydrogen peroxide is an oxidizing agent that oxidizes iodide ions to iodine in acidic medium:

$$2I^- + H_2O_2 + 2H^+ \longrightarrow I_{2(aq)} + 2H_2O$$

The reaction is monitored by adding a known volume of sodium thiosulphate solution and starch solution to the reaction mixture. The liberated iodine I_2 reacts with sodium thiosulphate and be reduced to iodide ions.

$$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$$

When thiosulphate ions are completely consumed, the liberated iodine I_2 reacts with starch solution and gives a blue color.

$$I_2 + \text{Starch} \longrightarrow \text{Blue complex}$$

The time elapsed before the appearance of blue color, gives an idea about the rate of the reaction.

$$\text{Rate} = \frac{[I_2]}{\text{time}} = 0.5\,\frac{[S_2O_3^{2-}]}{\text{time}}$$

Procedure

- Take four 250 mL conical flasks and label them as A, B, C and D.
- Add volume of 0.1 M potassium iodide (KI) solution to the flasks A, B, C and D respectively (as shown in table below).
- Add 10 mL of buffer acetate solution to each flask.
- Add 5 ml of starch solution to each flask.
- Add 10 ml of 0.05 M sodium thiosulphate ($Na_2S_2O_3$) solution to each flask.
- Complete the volume of the solutions to 100 mL by adding distilled water.
- Add volume of 3% H_2O_2 solution to flask A (as shown in table below), and start the stop-watch immediately.
- Stir the mixture using a glass rod and wait for the blue color to appear and record the time.
- Repeat the experiment for B, C and D flasks according to table, and record the time required in each case for the blue color first appears.
- Repeat what you are doing to flask A, but at temperature higher than the room temperature (50-60)°C or at 0°C.

Flask	0.1 M KI solution (ml)	Buffer solution (ml)	Starch solution (ml)	0.05 M $Na_2S_2O_3$ (ml)	Water (ml)	3% H_2O_2 (ml)
A	10	5	5	5	70	5
B	5	5	5	5	75	5
C	5	5	5	5	70	10
D	15	5	5	5	65	5
A at (T) C^o	10	5	5	5	70	5

Data and calculation:

Part A – Finding the order of each reactant (n, m, p) and then the Rate Law using the method of initial rates. Show a sample calculation. *(Hint : $M_i \times V_i = M_f \times V_f$)*

Flask	$[I^-]$ if 0.1 M KI solution was used	$[H_2O_2]$ if 3% H_2O_2 used	$[S_2O_3^{2-}]$ if 0.05 M $Na_2S_2O_3$ used	Time (t) for the appearance of blue color (s)	Rate $= \frac{½[S_2O_3^{2-}]}{t}$	K
A						
B						
C						
D						
A at ( )						

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Part B – The Effect of Temperature on the Reaction Rate

In this part we will use the temperature data collected for part-A in first run and final run to determine the value of the activation energy, Ea for the reaction. (Hint: find the rate constant k_2 at T = ……. ^{o}C)

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